

Communicating with Data – Embellishment

Dr. Ab Mosca (they/them)

Slides based off slides courtesy of Jordan Crouser (<https://jcrouser.github.io/>)

Recall

- What is one caveat to keep in mind when encoding data via color (hue, or saturation)?

Plan for Today

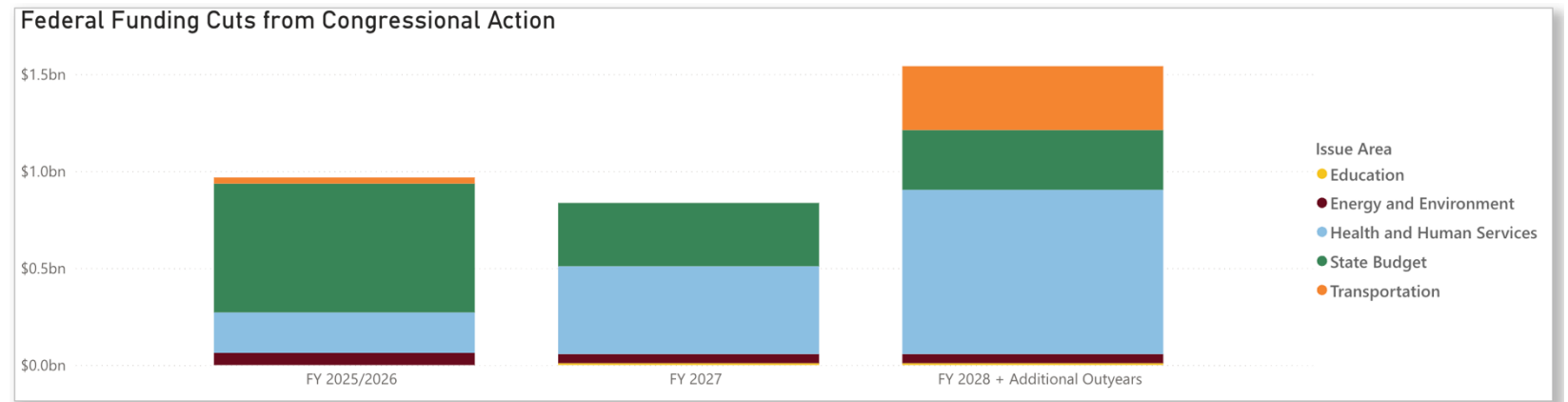
- What is embellishment for visualization?
- Chart Junk
- Definition
- Benefits and Drawbacks

Brainstorm

- What do you think embellishment for a visualization is?

Brainstorm

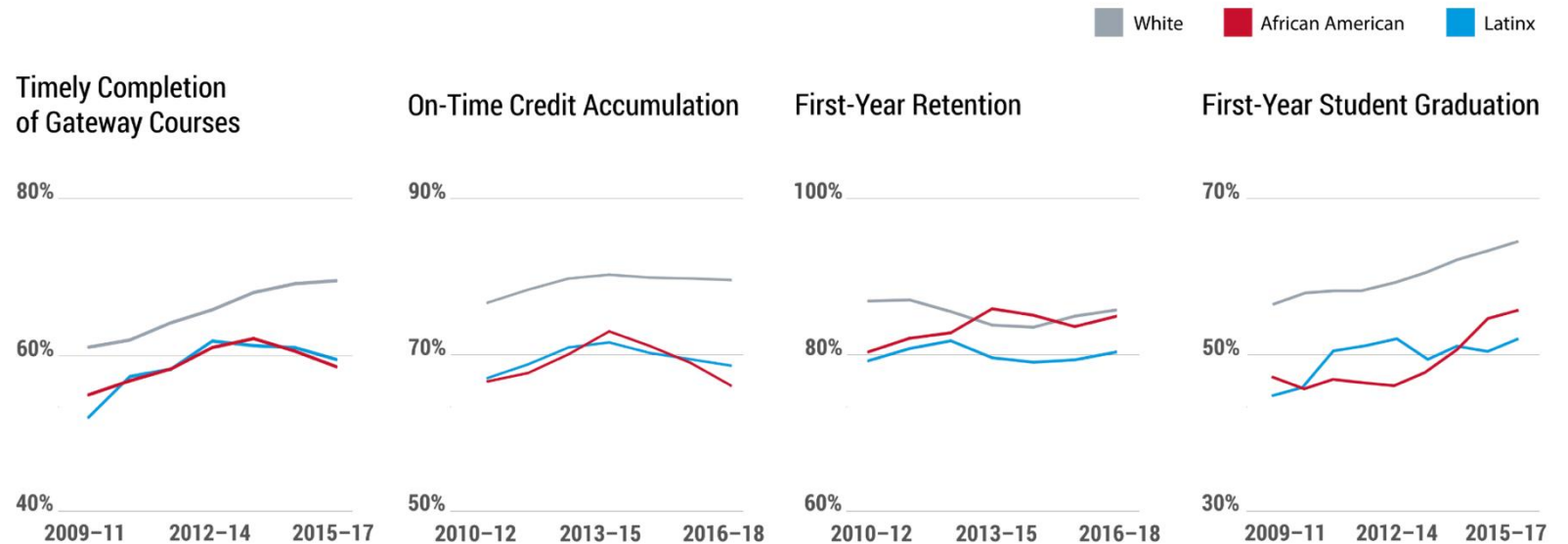
- How might you embellish these visualizations?



<https://www.mass.gov/info-details/impact-of-trump-administration-and-congressional-cuts-on-massachusetts>

State University Student Success Metrics: Gaps by Race/Ethnicity

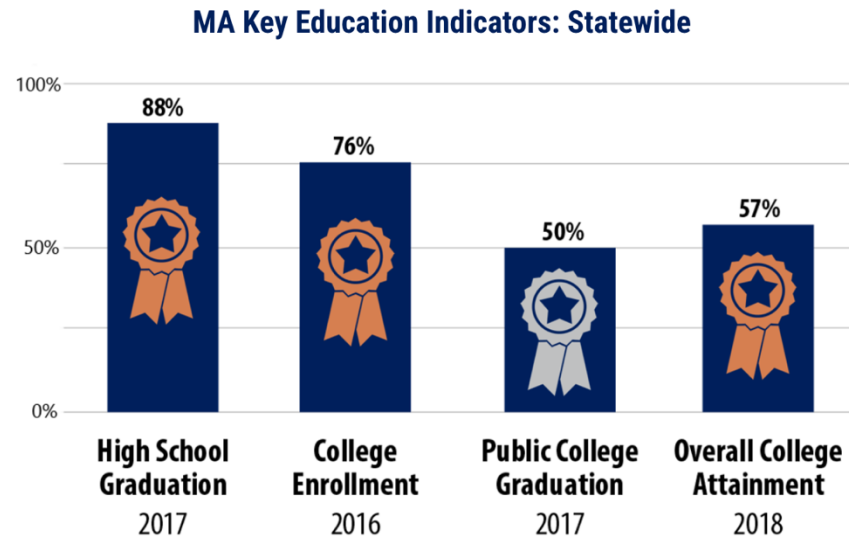
Average Rate among State Universities



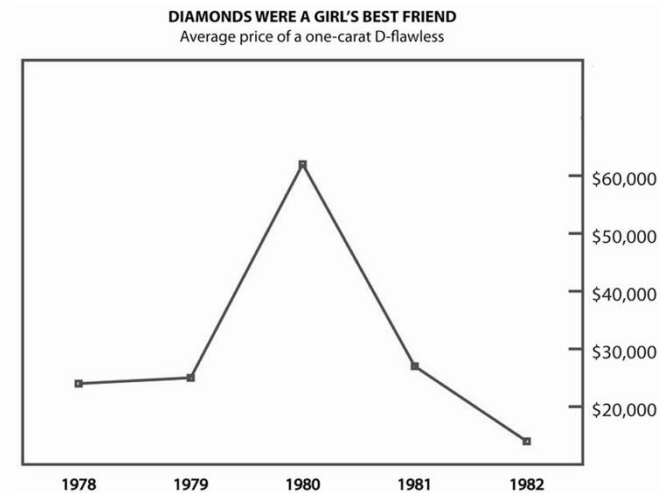
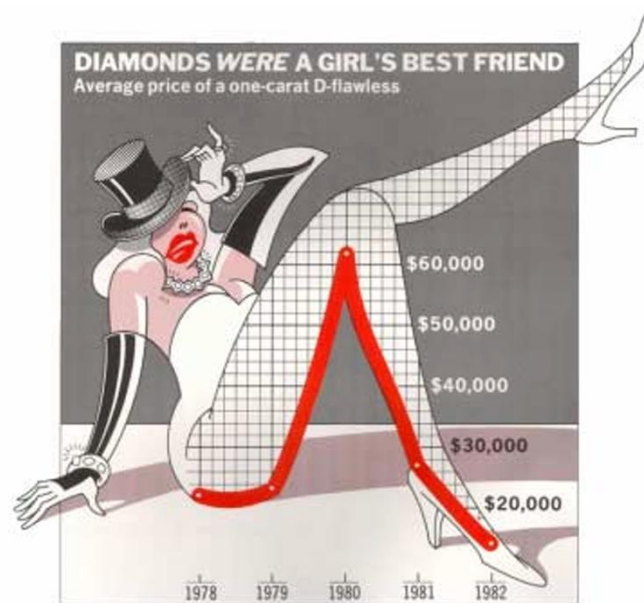
<https://www.mass.edu/datacenter/pmars/reports/2019/Success-Completion.asp>

Brainstorm

- Are these embellished? Why or why not?



<https://www.mass.edu/datacenter/equityspotlight-latinomales.asp>



Scott Bateman, Regan L. Mandryk, Carl Gutwin, Aaron Genest, David McDine, and Christopher Brooks. 2010. Useful junk? the effects of visual embellishment on comprehension and memorability of charts. <https://doi.org/10.1145/1753326.1753716>

Chart Junk

Edward Tufte

- **Data-Ink** == Ink on the page that displays data
- **Chart Junk** ==
 - Non-data ink
 - Purely style elements

Chart Junk

Redraw the following *without* chart junk



Chart Junk

Redraw the following *without* chart junk

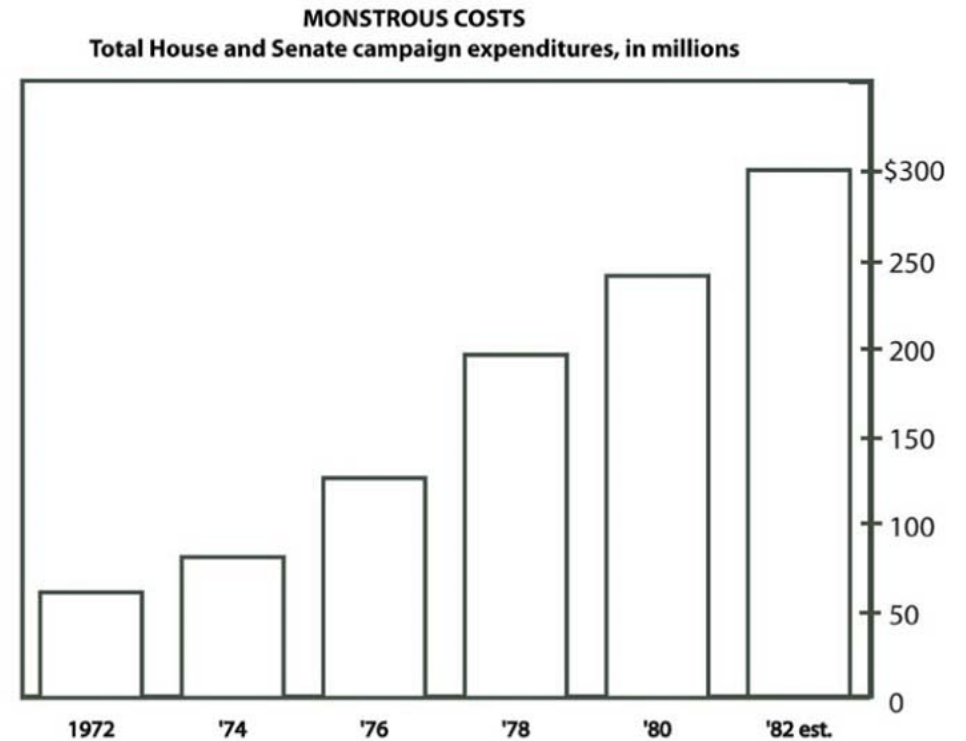
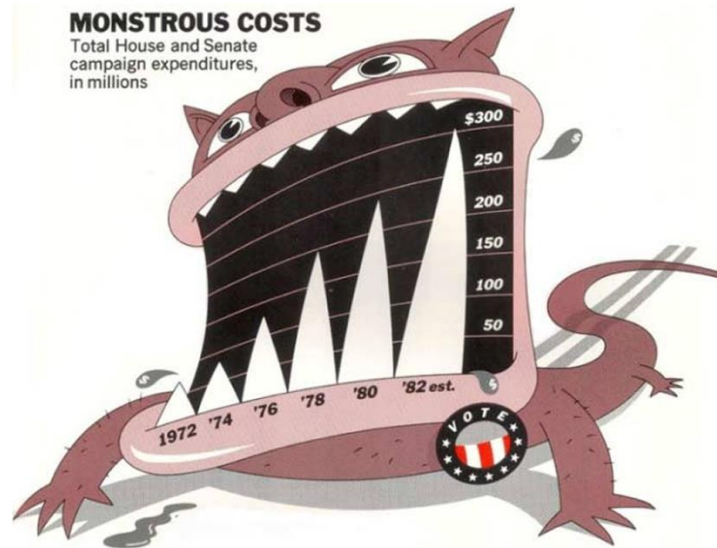


Chart Junk

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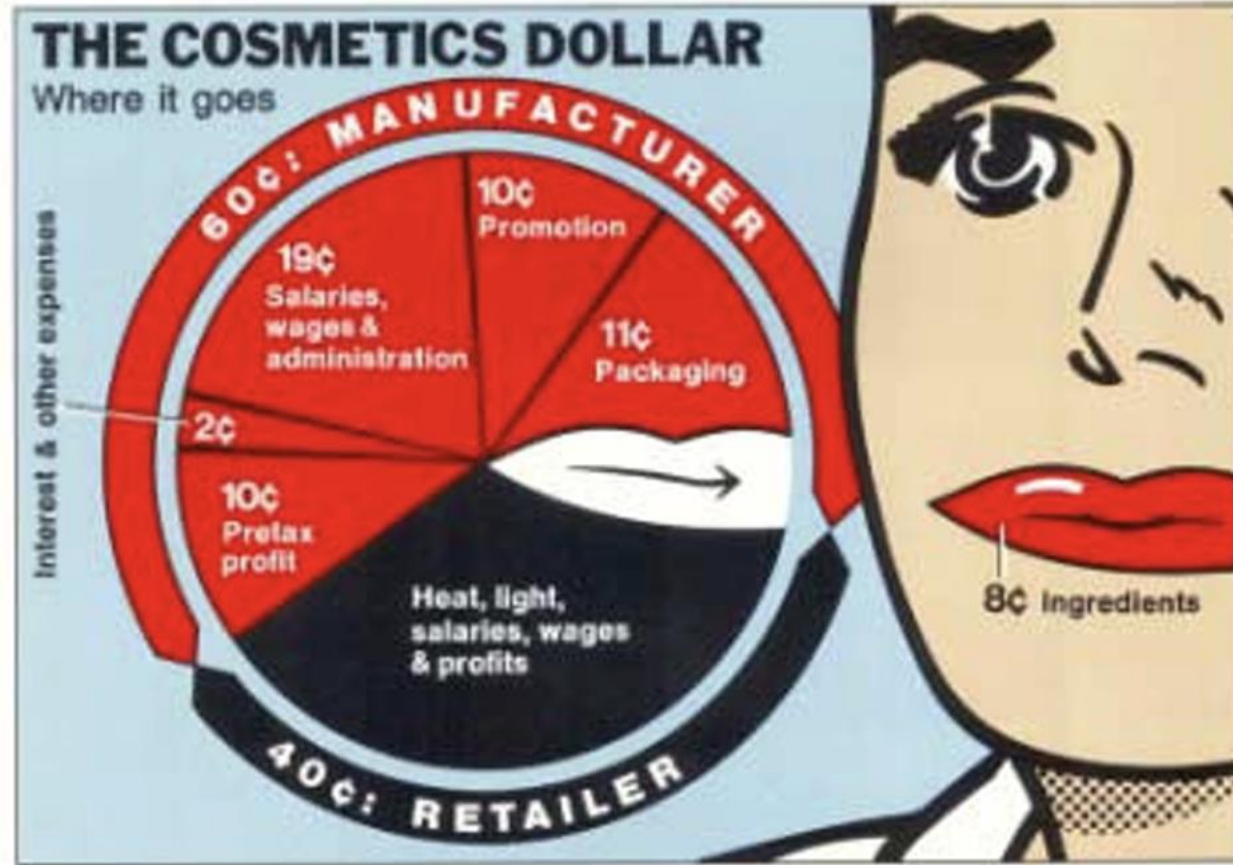


Chart Junk

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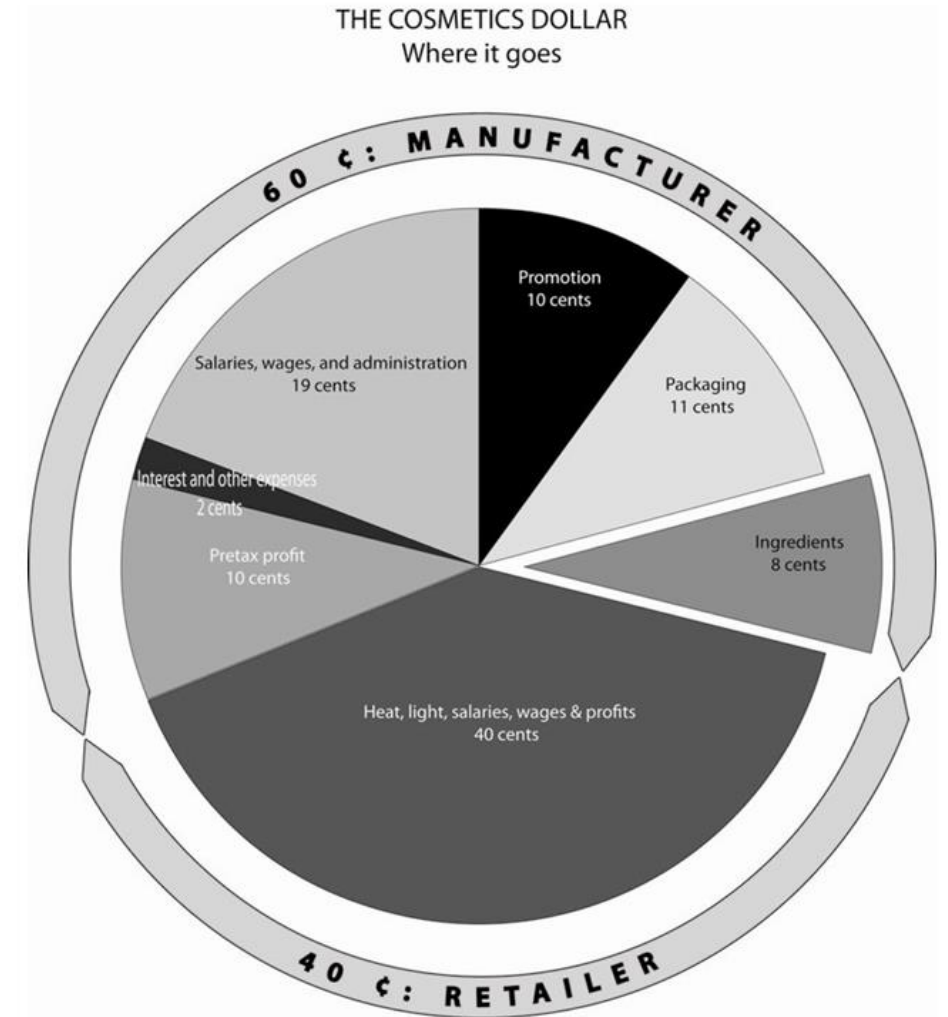
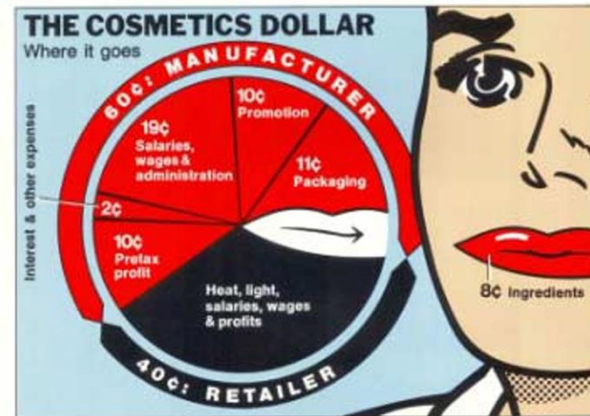


Chart Junk

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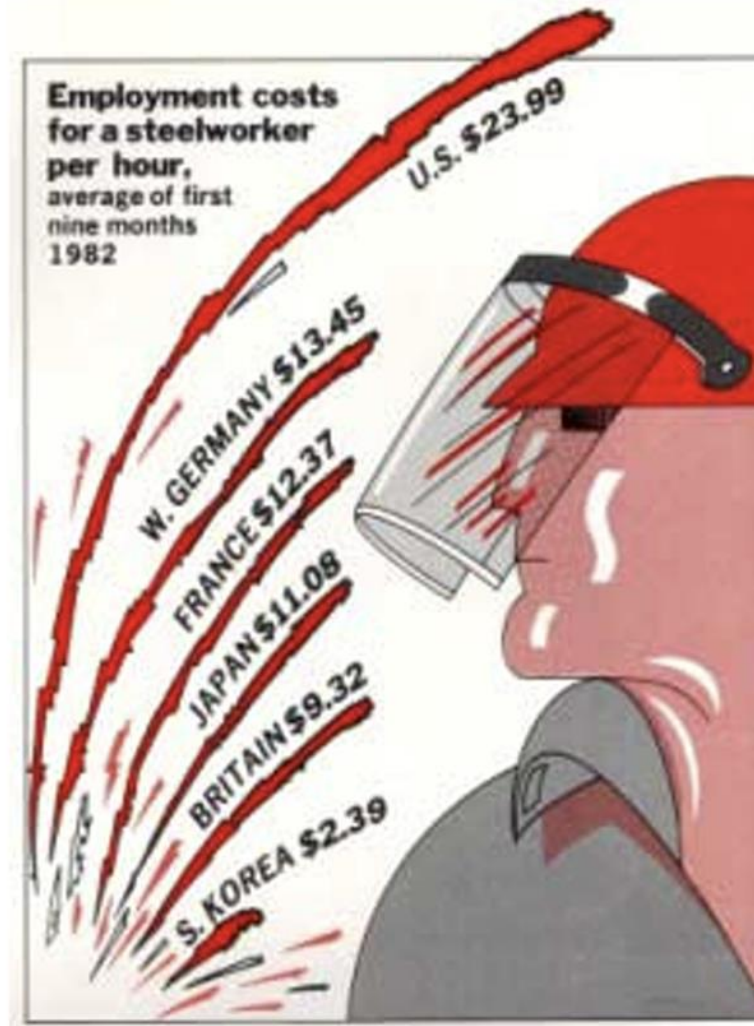


Chart Junk

Redraw the following *without* chart junk

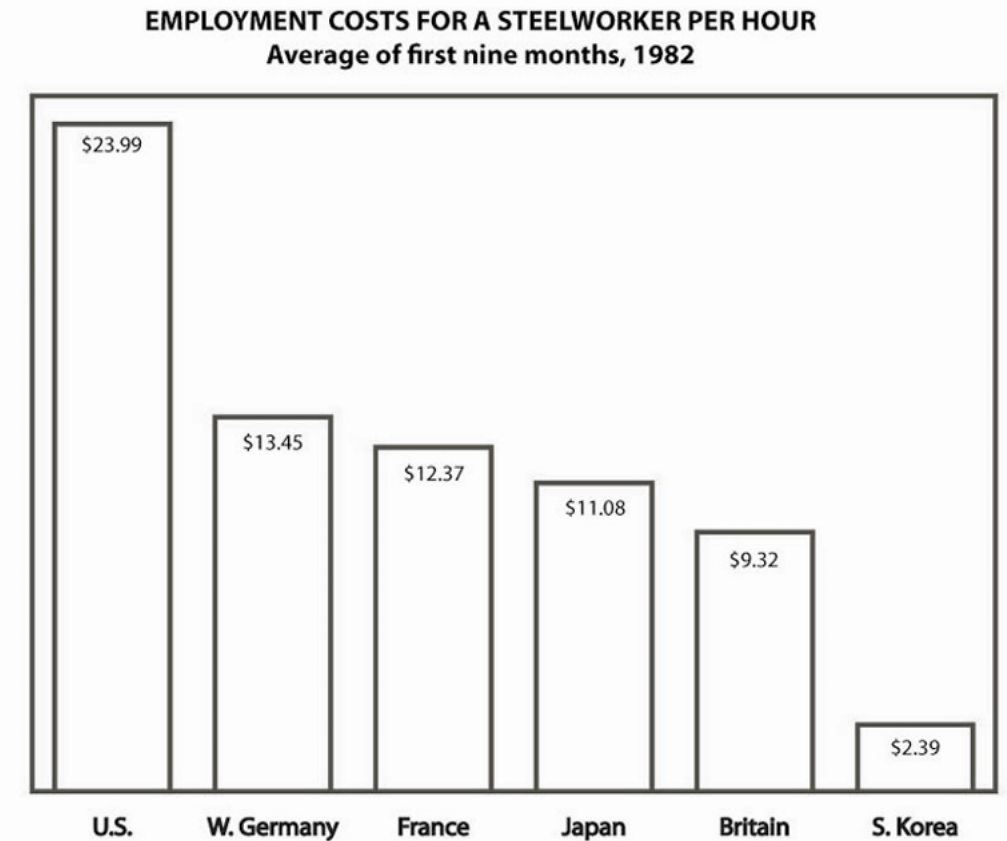
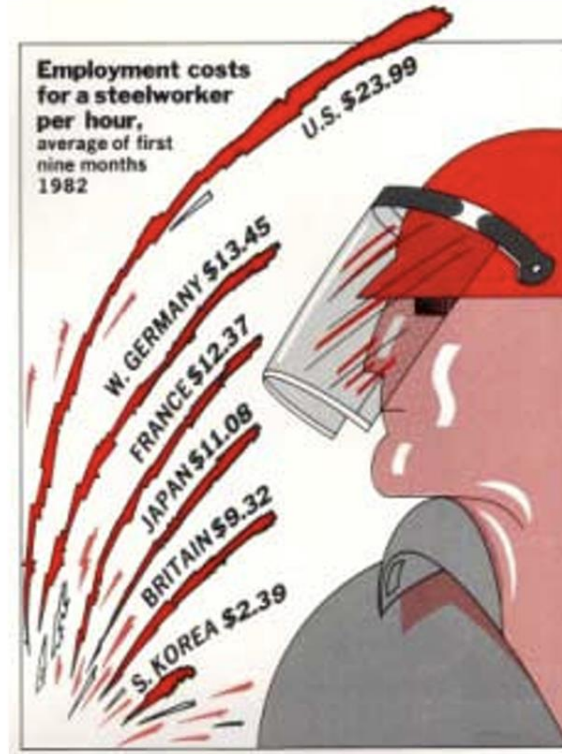


Chart Junk

Edward Tufte

- **Data-Ink** == Ink on the page that displays data
- **Chart Junk** ==
 - Non-data ink
 - Purely style elements
- Tufte's take: the best visualizations minimize chart junk and focus explicitly on representing data
- **Do you agree?**

Definition

When we say **embellishment** in a visualization context, we're referring to graphical elements that do not directly display data, or that are added for stylistic reasons to a chart.

You can think of embellishment as chart junk. Like anything, there are benefits and drawbacks...

Embellishment and Recall / Memorability

- Short-term recall of chart subject, categories shown, and trend appears unaffected by embellishment
- Long-term recall of chart subject, categories shown, and trend appear affected by embellishment

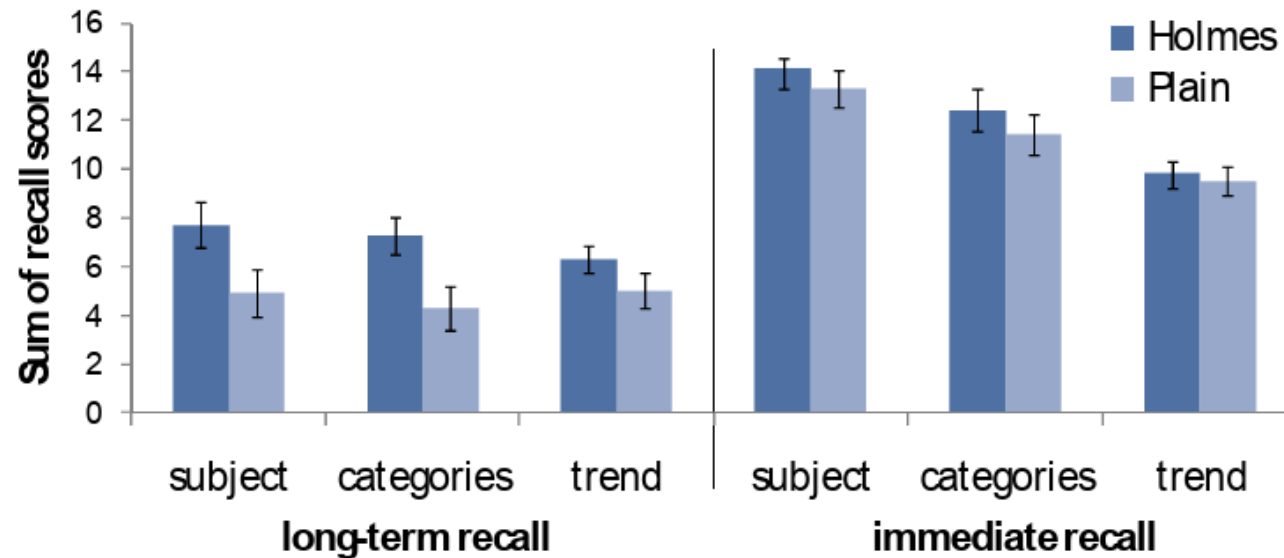


Figure 5. Means $\pm$ S E for recall scores for long-term and immediate recall.

Embellishment and Recall / Memorability

- Short- and long-term recall of a chart's value message are affected by embellishment

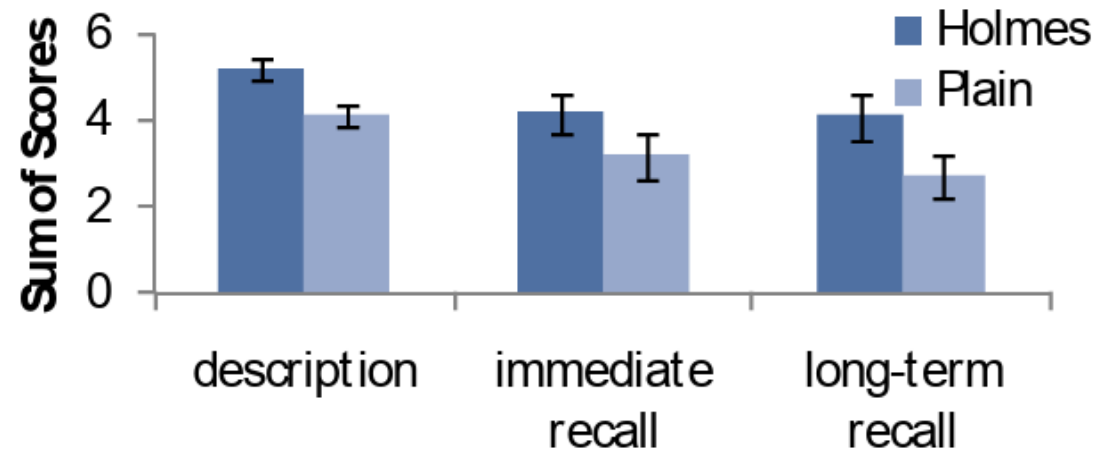


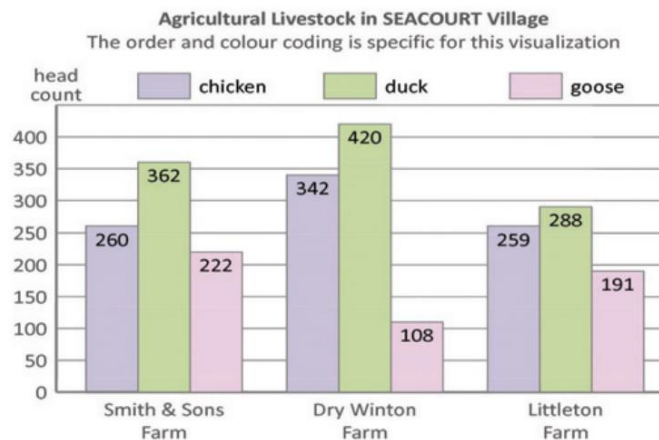
Figure 6. Means $\pm$ SE for sum of value message scores.

Embellishment and Recall / Memorability

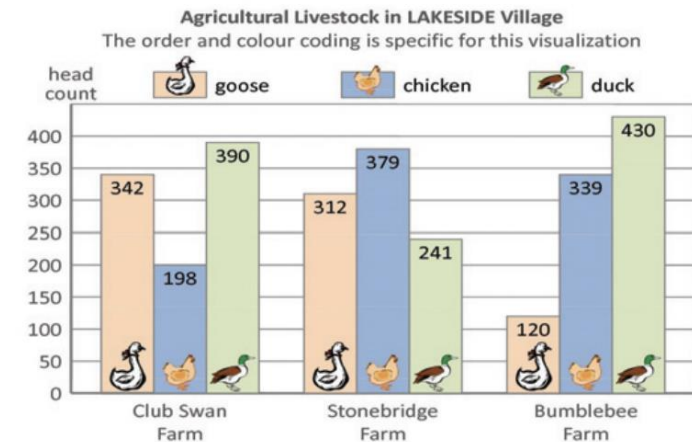
- What readers recall about visualizations is not hindered by pictograms. In other words, with pictograms the message of the visualization is correctly recalled (and in fact, often better)
- However, pictograms can hurt the understanding of a visualization. Any time people spend on a pictogram is time not spent on the data or the message (unless the pictogram contains part of the data or message), and if the link between the data/message and the pictogram is not clear, people may misunderstand and/or misrecall what the visualization was about

Embellishment and Cognition

- When visual embellishments relate to information to be remembered (ex. groups), they improve working memory
- In other words, visual embellishments can help free up cognitive space for *other* tasks



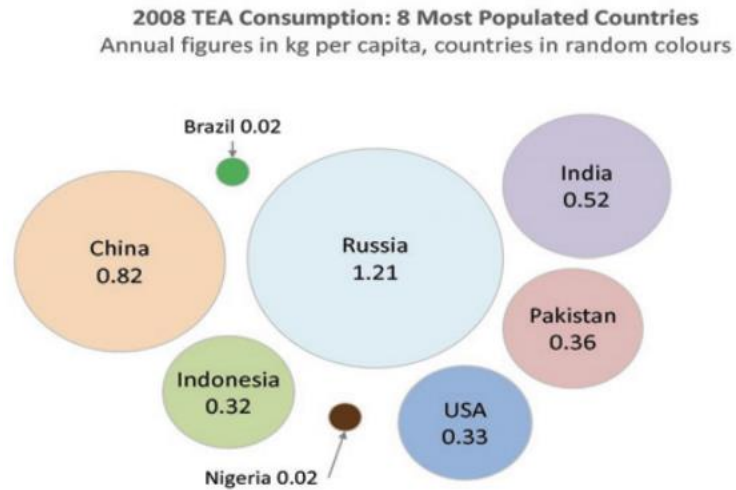
(m) VS Stimulus 3.3



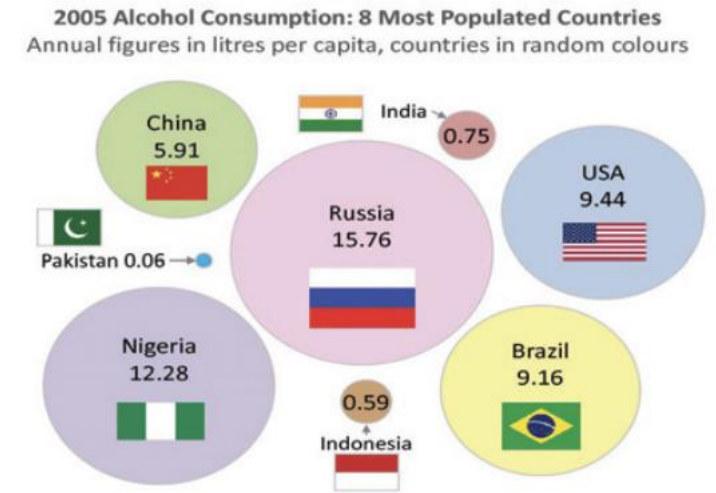
(n) VS Stimulus 3.4m

Embellishment and Cognition

- Visual embellishment can negatively impact visual search, by distracting from the search target



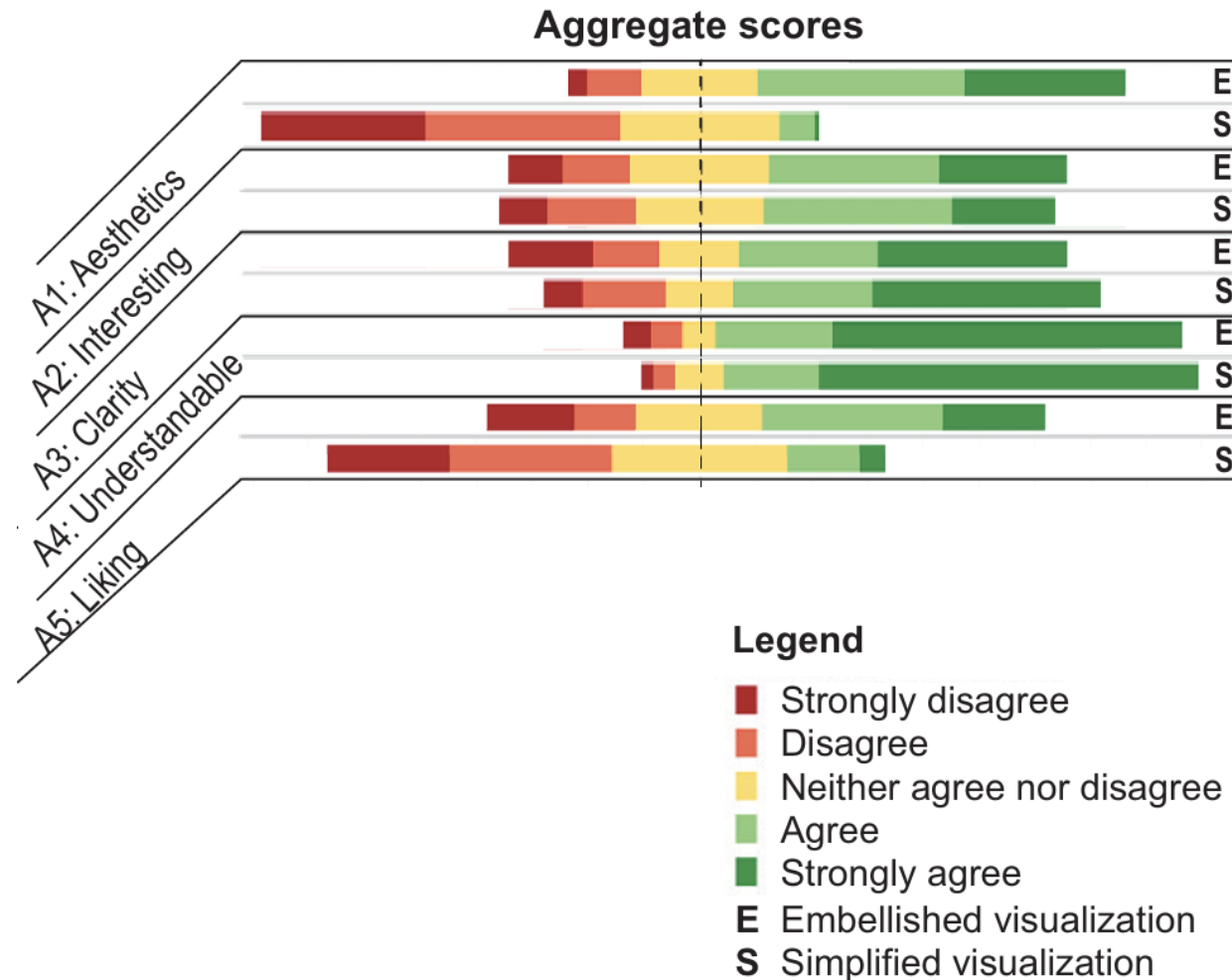
(q) VS Stimulus 2.1



(r) VS Stimulus 2.2m

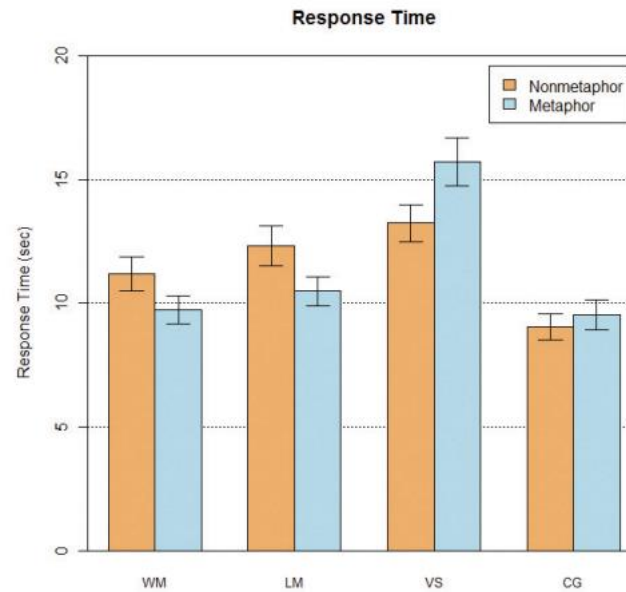
Embellishment and Reader Preference

- Readers tend to rate embellished charts as preferable

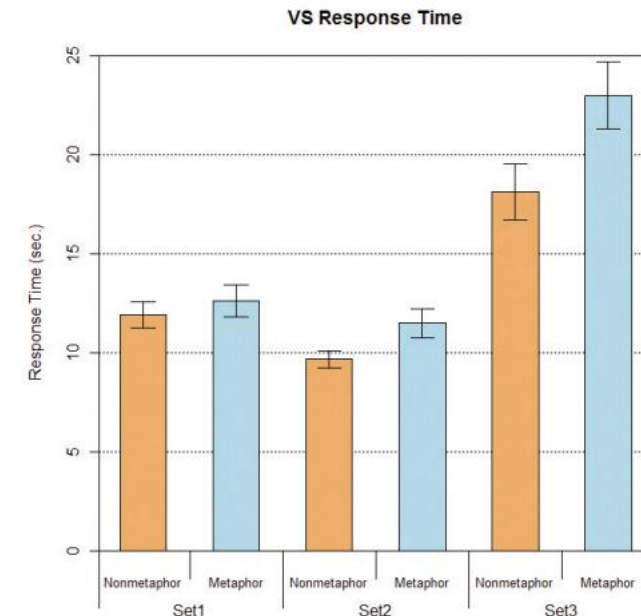


Embellishment and Time

- Embellishment affects response time



(b). Response Time.

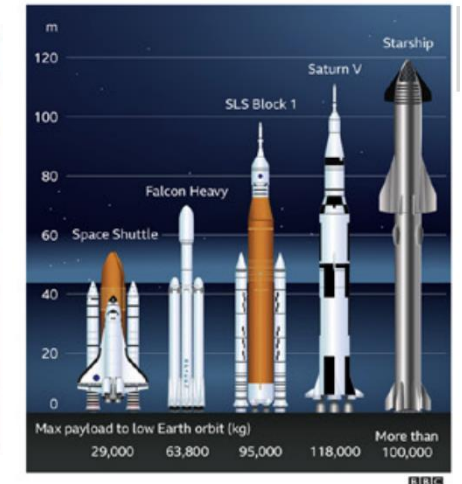


(c). VS Response Time.

When might longer response time be positive vs. negative?

Embellishment and Engagement

- When engagement is measured via insights, embellished and simple charts are found to have similar levels
- Interviews show embellishment is associated with enticement, and people like visualizations that look “picture-like”



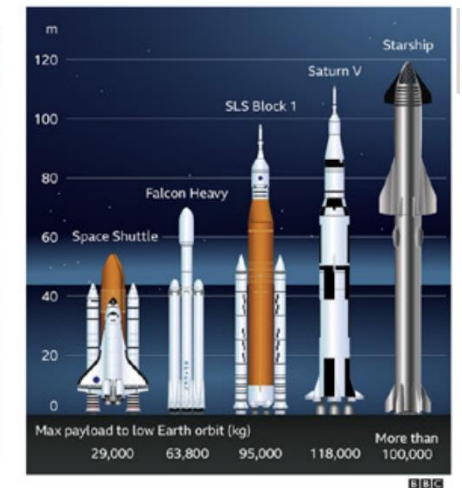
M. Alebri, E. Costanza, G. Panagiotidou and D. P. Brumby, "Embellishments Revisited: Perceptions of Embellished Visualisations Through the Viewer's Lens" in IEEE TVCG, vol. 30, no. 01, pp. 1424-1434, Jan. 2024, doi: 10.1109/TVCG.2023.3326914.

V. Braun and Victoria Clarke. Thematic Analysis A Practical Guide. Sage, first ed., 2021. doi: 10.1177/1035719X21105825

A. Vande Moere, M. Tomitsch, C. Wimmer, B. Christoph, and T. Grechenig. Evaluating the Effect of Style in Information Visualization. IEEE TVCG, 18(12):2739– 2748, 2012. doi: 10.1109/TVCG.2012.221

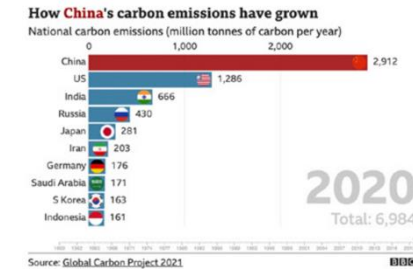
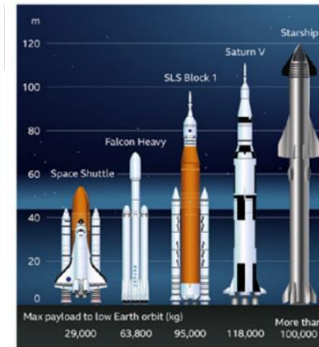
Embellishment and Seriousness

- Some viewers feel that embellished visualizations underestimate their ability to read a visualization and are patronizing



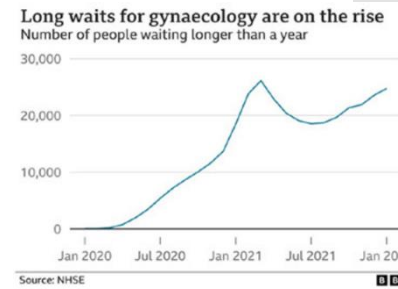
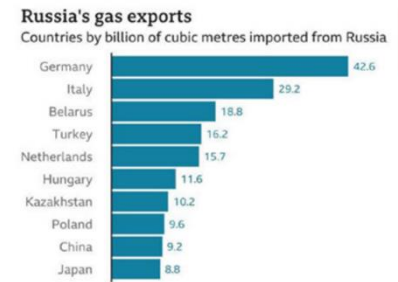
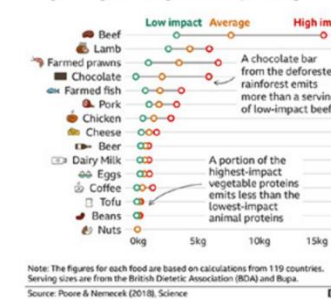
Embellishment and Trust

- Embellishment can erode trust; some viewers associate it with tabloid or sensational reporting



Beef has the biggest carbon footprint – but the same food can have a range of impacts

Kilograms of greenhouse gas emissions per serving

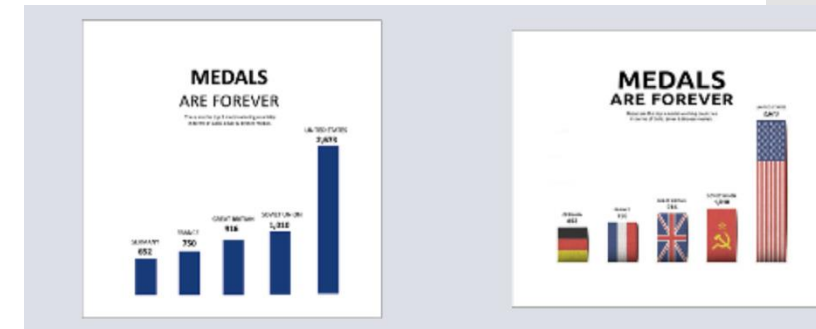


Less Trustworthy

More Trustworthy

Embellishment and Trust

- Comic-style visualizations perceived as less credible
- Image-like visualizations were perceived as similarly credible compared to standard visualizations.
- Plain bars in a bar chart perceived as more credible than cartoon-style bars, while both plain bars and image bars were perceived as similarly credible



Less Trustworthy

More Trustworthy

Your Turn!

- Choose some subset of data from the Smith common dataset: <https://www.smith.edu/your-campus/offices-services/institutional-research/data-about-smith>
- Draw a plain / simple visualization
- Add embellishment to the visualization that you think will add memorability so prospective students remember Smith and apply
- Add embellishment to the visualization that you think will add memorability so readers remember specific data points from the visualization